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Letter

Nonequilibrium Work Relations and Response Theories in Ensemble Quantum Systems

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ABSTRACT: We develop a nonequilibrium response theory for macroscopic quantum systems that separates the contributions of ensemble heterogeneity and intrinsic quantum uncertainty. To accomplish this, we describe systems with a quantum *P*-ensemble, which goes beyond the standard density matrix description by explicitly specifying the classical heterogeneity between individual quantum systems in an ensemble. We use the *P*-ensemble formalism to present quantum generalizations of linear response theory and the Jarzynski nonequilibrium work relation. We derive these generalizations from a Bochkov-Kuzovlev generating functional for quantum *P*-ensembles, which can be further utilized to derive all orders of response theory that apply to ensemble quantum systems. We contrast these developments with their ρ-ensemble analogs, and we discuss how these *P*-ensemble theories provide a guide for an effective application of single molecule experiments.

$$\overline{\left(e^{\int_0^\tau dt u(t)\boldsymbol{A}(\rho)_F(t)} e^{-\beta \boldsymbol{H}(\rho)_F(\tau)} e^{\beta \boldsymbol{H}(\rho)_F(0)}\right)} = e^{-\beta\Delta f}\,\overline{\left(e^{\epsilon_A \int_0^\tau \tilde{u}(t)\boldsymbol{A}(\rho)_R(t)}\right)},$$

Much of our success in developing technologies that deliberately harness the unique properties of quantum physics, including quantum computing,[1] sensing,[2] and simulation,[3] depends on our ability to design the dynamics of quantum systems that are out of equilibrium. In ensemble systems, design efforts must account for the unavoidable influence of environmental heterogeneity, which causes the dynamics of ensemble members to differ from one another. The key to understanding these differences and their effect on the observed system dynamics is to relate them to the microscopic fluctuations of the individual ensemble members. The challenge with formulating this relationship in quantum systems is that their microscopic fluctuations do not arise solely through the sampling of state space, as they do in classical systems, but also incorporate the effects of intrinsic uncertainty (e.g., the factors that determine probability wave function collapse). These effects tend to muddle the interpretation of standard nonequilibrium theories and have thus hindered progress in the study of nonequilibrium quantum dynamics.

The theoretical basis for analyzing and interpreting the dynamics of nonequilibrium systems requires the ability to specify a system's microscopic thermal fluctuations at equilibrium. The relationships between these microscopic fluctuations and macroscopic system properties are described by well-established theoretical principles, such as the Green-Kubo theory,[4−6] the fluctuation−dissipation theorem,[7−12] and nonequilibrium work relations.[13−19] These theories formalize how changes in ensemble statistics arise due to variations in the response and evolution of ensemble members under the influence of external forces. Past efforts to extend these theories to quantum systems have addressed the issue of quantum uncertainty through the use of density matrices. For example, quantum generalizations of the Jarzynki Equality and the Crooks Fluctuation theorem have been derived by defining energy and work in terms of density matrix measurements before and after a work protocol is applied.[20,21] While these efforts have enabled a statistical interpretation of non-equilibrium dynamics, they are challenging to analyze because the density matrix convolves the statistical influences of ensemble heterogeneity and probabilitic quantum uncertaintly.

In this letter, we present a quantum response theory and nonequilibrium work relation that address this challenge by characterizing how intrinsic uncertainty and ensemble heterogeneity separately contribute to microscopic thermal fluctuations. Separating these contributions reveals how different forms of uncertainty are encoded in nonequilibrium response and provides a practical set of tools for interpreting emerging single molecule experiments.[22−25] Our approach builds upon the state space formalism presented in ref 26 that provides a classical-like characterization of the dynamical flow of quantum ensembles. We use this formalism to derive a novel Bochkov-Kuzovlev (BK) generating functional[18] that in turn provides expressions of fundamental statistical mechanical equations, including linear and nonlinear response theories of all orders and the nonequilibrium work relations.

We begin by outlining frameworks for defining the quantum ensembles and correlation functions that characterize the dynamics of individual ensemble members. We then use these definitions to formulate a BK generating functional. Using this

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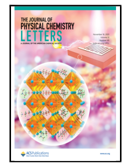

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generating functional, we derive a new nonequilibrium work relation and linear response theory for quantum ensembles. We then contrast these theorems to their preexisting quantum and classical counterparts,[16,18,19] and provide an explicit example comparing their applications in a simple spin system.

Defining Quantum Ensembles and Correlation Functions. Quantum systems are typically defined in the ρ-ensemble, where the state of the entire quantum ensemble is specified by a single density matrix, $\boldsymbol{\rho}$. The density matrix encodes all ensemble observables, $\mathbf{A}$, through the expectation formula, $\overline{A}[\boldsymbol{\rho}] \equiv \mathrm{Tr}\{\mathbf{A}\boldsymbol{\rho}\}$, and can thus provide insight into the statistics of macroscopic observables. However, despite the obvious utility of this formalism, the ρ-ensemble lacks the information necessary to distinguish the statistical properties of individual microscopic systems in observationally equivalent ensembles.[27] For example, members of an ensemble of two-state systems that are all in the same mixed state, $(|0\rangle\langle 0| + |1\rangle\langle 1|)/2$, will exhibit different statistics from those of an ensemble of equal populations of the pure states $|0\rangle\langle 0|$ and $|1\rangle\langle 1|$, despite the fact that these ensembles behave identically under macroscopic observation (i.e., are described by the same ensemble density matrix $\boldsymbol{\rho}$). In particular, the ρ-ensemble description does not indicate whether the systems are all in an identical quantum state as in the first ensemble or if different systems are found in different states as in the second ensemble.

The statistical response of individual microscopic subsystems can be measured by certain single molecule experiments.[22−25] Repeated application of such experiments have the potential to enable the effective separation of ensemble heterogeneity (i.e., differences between individual systems) and intrinsic quantum uncertainty (e.g., probabilistic variations in wave function collapse upon observation). The physical implications of this separation cannot be fully understood in the context of the ρ-ensemble. To formalize this separation it is necessary to explicitly define the distribution that characterizes ensemble heterogeneity.

This distribution is defined by the P-ensemble, in which the state of each microscopic system (ensemble member) is expressed as a density operator, $\boldsymbol{\gamma}$. Since $\boldsymbol{\gamma}$ is a single system state, its statistical properties relate only the system's intrinsic quantum uncertainty. The distribution of states within the ensemble is then defined by $P(\boldsymbol{\gamma})$, a classical probability distribution over Liouville space.[28] The ρ-ensemble can be derived from the P-ensemble through the following average,

$$\boldsymbol{\rho} = \int \boldsymbol{\gamma} P(\boldsymbol{\gamma})\, \mathrm{d}\boldsymbol{\gamma} \tag{1}$$

and hence we refer to $\boldsymbol{\rho}$ as the *ensemble density operator*. Moreover, each $\boldsymbol{\gamma}$ is a density operator and can thus alternately define a ρ-ensemble or -subensemble. The $\boldsymbol{\gamma}$ integral in eq 1 is evaluated over the set of all possible system density matrices $\boldsymbol{\gamma}$. To evaluate this integral for a D dimensional system, we can express $\boldsymbol{\gamma} = \sum \gamma_{ij}\, |i\rangle\langle j|$ in the D^2 dimensional orthonormal matrix basis $|i\rangle\langle j|$. The integral over the set of density matrices can then be written as an integral over the manifold of complex valued matrix elements γ_{ij} that satisfy the following constraints: (1) γ_{ii} are real, (2) $\boldsymbol{\gamma}$ is normalized, that is, $\sum_i \gamma_{ii} = 1$, (3) $\boldsymbol{\gamma}$ is Hermitian, that is, $\gamma_{ij} = \gamma_{ji}^*$, and (4) there is a coherence constraint expressed as $|\gamma_{ij}|^2 \leq \gamma_{ii}\gamma_{jj}$. In general, evaluating this integral is no more complicated than evaluating an integral of a vector valued function over an N-dimensional complex manifold or alternatively over a $2N$-dimensional real manifold. For example, below we consider a 2-level spin system where the integral is simply performed over the Bloch sphere.

Equation 1 clearly shows that for every ρ-ensemble description, there are infinitely many P-ensemble descriptions. This ambiguity of the ρ-ensemble is a well-known property of mixed-state density matrices, which in general can represent an infinite number of statistical mixtures of pure states in addition to improper mixtures that arise due to the formation of system-bath entanglement. Each of these mixtures, represented by a given choice of $P(\boldsymbol{\gamma})$, is a different interpretation of the mixed state thus implying a different physical mechanism generating the observed statistics. Representing the state of a quantum ensemble as a P-ensemble therefore provides the conceptual advantage of unambiguously describing a specific physical interpretation to the ensemble statistics encoded in $\boldsymbol{\rho}$. As discussed in ref 26, this relationship between P- and ρ-ensembles also pertains to the Hilbert space distributions used in stochastic unraveling methods[29−33] that represent continuously observed quantum systems.[33−35] These works are built on the mathematical foundation of Davies' quantum stochastic processes,[36] as described in detail in ref 27.

Information contained within the P-ensemble allows for the definition of single-system averages and correlation functions that are essential ingredients of statistical mechanical response theories. The general form of a ρ-ensemble quantum correlation function (QCF) for observables A and B separated by time t is given by

$$c_{AB}(t)[\boldsymbol{\rho}] = \overline{A(t)B(0)}[\boldsymbol{\rho}] - \overline{A}(t)[\boldsymbol{\rho}]\overline{B}(0)[\boldsymbol{\rho}] \tag{2}$$

where $\overline{A}(t)\,[\boldsymbol{\rho}] \equiv \mathrm{Tr}\{\mathbf{A}_{\mathrm{H}}(t)\boldsymbol{\rho}_{\mathrm{H}}\}$ denotes the quantum average of observable $\mathbf{A}$ at time t and

$$\overline{A(t)B(0)}[\boldsymbol{\rho}] \equiv \mathrm{Tr}\{\mathbf{A}_{\mathrm{H}}(t)\mathbf{B}_{\mathrm{H}}(0)\boldsymbol{\rho}_{\mathrm{H}}\}$$

where the subscript H denotes the use of the Heisenberg picture. In the P-ensemble this correlation function can be expressed as a sum of two separately observable single-system correlation functions,

$$c_{AB}(t)[\boldsymbol{\rho}] = \langle c_{AB}(t)[\boldsymbol{\gamma}]\rangle_P + C_{\overline{A}\overline{B}}(t)[P] \tag{3}$$

The first term in eq 3 evaluates the quantum correlation function of each ρ-subensemble, $\boldsymbol{\gamma}$, using eq 2. We then take the classical average of these quantities weighted by $P(\boldsymbol{\gamma})$ as denoted by the angle brackets $\langle x\rangle_P \equiv \int x[\boldsymbol{\gamma}]P(\boldsymbol{\gamma})\, \mathrm{d}\boldsymbol{\gamma}$. This averaged quantum correlation function can therefore be explicitly written as

$$\langle c_{AB}(t)[\boldsymbol{\gamma}]\rangle_P \coloneqq \int P(\boldsymbol{\gamma})(\overline{A(t)B(0)}[\boldsymbol{\gamma}] - \overline{A}(t)[\boldsymbol{\gamma}]\overline{B}(0)[\boldsymbol{\gamma}])\, \mathrm{d}\boldsymbol{\gamma} \tag{4}$$

The classical correlation function (CCF), $C_{\overline{A}\overline{B}}(t)$, of the quantum averaged observables is given by

$$C_{\overline{A}\overline{B}}(t)[P] \equiv \langle \overline{A}(t)[\boldsymbol{\gamma}]\overline{B}(0)[\boldsymbol{\gamma}]\rangle_P - \langle \overline{A}(t)[\boldsymbol{\gamma}]\rangle_P \langle \overline{B}(0)[\boldsymbol{\gamma}]\rangle_P \tag{5}$$

The details of this derivation can be found in Chapter 5 of ref 27. The relative weight of these two correlation functions quantifies the relative contribution of intrinsic single-system uncertainty and ensemble heterogeneity and thus generally differs between ensembles.

To gain insight into the physical meaning of the P-ensemble correlation function, consider some limiting cases. In the case where all systems in the ensemble are initialized in an identical state, that is, $P(\boldsymbol{\gamma}) = \delta(\rho-\gamma)$, the second (CCF) contribution in

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eq 3 vanishes, and all fluctuations arise from the intrinsic uncertainty of the individual quantum system, captured by the QCF. In contrast, the first (QCF) term will vanish in the limit that the observables A and B are uncorrelated at the single system level. They may however be correlated at the ensemble level, leading to nonzero CCF, if the method of preparing systems in the ensemble correlates the average value of the two operators.

It is possible, at least in principle, to separately account for the intrinsic and ensemble contributions to the QCF. Similarly to the ensemble QCF, insight can be gained into the P-ensemble correlation function, $c_{AB}(t)$ $[\boldsymbol{\gamma}]$, by repeatedly initializing and probing a single quantum system (e.g., via ultrafast single molecule spectroscopy). Similarly, the ensemble CCF, $C_{\overline{AB}}(t)[P]$, can be measured by resolving the time dependent quantum averages of the observable such as $\overline{A}(t)$. In practice, however, these experiments present significant technical challenges that have yet to be overcome in most systems. Current techniques are often limited by repetition rate and thus tend to average over the fastest degrees of freedom. Such approaches would therefore provide an incomplete yet still useful separation of the effects of ensemble heterogeneity and intrinsic uncertainty.

The relationship of ρ-ensemble QCFs to system response have been well studied and described by Kubo theory near equilibrium,[4] and the quantum Fluctuation Theorems far from equilibrium.[37−39] These, theorems also characterize the QCFs for the (generally nonequilibrium) $\boldsymbol{\gamma}$'s since they each can alternately define a $\boldsymbol{\rho}$ ensemble. However, the relationships between the CCFs and macroscopic observables remain uncharacterized and will thus be the primary focus of this letter. We will show below that by using the properties of the state space distribution formalism, P ensemble CCFs follow response and nonequilibrium work relations analogous to those derived in classical thermodynamics.

Equilibrium Distributions. Most statistical mechanical theories are built upon the foundational assumption that the probability of observing a given microstate, ν, within the canonical ensemble is proportional to exp $[-\beta E_\nu]$, where E_ν is the microstate energy, $\beta = 1/k_{\mathrm{B}}T$, and $k_{\mathrm{B}}T$ is the Boltzmann constant times temperature. We formulate an analog of this distribution law for the P-ensemble starting with the maximum entropy principle.[40,41] Specifically, we define the classical entropy of a given P-ensemble as,

$$S_P \equiv -\int \mathrm{d}\boldsymbol{\gamma} P(\boldsymbol{\gamma}) \log P(\boldsymbol{\gamma}) \tag{6}$$

where the integral is taken over all possible states. Typically, this integral is taken over the set of all density matrices, as in eq 1, however, when modeling some ensembles, the systems may have to obey certain physical constraints that correspondingly restrict the integral to a subset of states. A common example of such a constraint, which will appear in the spin example below, arises when considering the distribution of states for an ensemble of isolated single systems. In this closed system case, there is no bath for the systems to form entanglements with, therefore each system must be found in a pure state, restricting the set of possible states to the manifold of pure state density matrices (i.e., $|\gamma_{ij}|^2 = \gamma_{ii}\gamma_{jj}$).

We use the method of Lagrange multipliers to determine the distribution that maximizes this entropy subject to a Gibbs constraint on the mean of the distribution, $\boldsymbol{\rho}$. This constraint ensures that $\boldsymbol{\rho}$ shows no coherences between energy eigenstates at equilibrium, and that each eigenstate is occupied with Boltzmann probability $\sim e^{-\beta E_n}$. This procedure, as described in detail within the SI, yields a Boltzmann-like equilibrium distribution for a system with Hamiltonian $\mathbf{H}(\lambda; \mathcal{B})$,

$$P_\lambda(\boldsymbol{\gamma}; \mathcal{B}) = \frac{\mathrm{e}^{-\beta' \bar{H}[\boldsymbol{\gamma}](\lambda; \mathcal{B})}}{\mathcal{Z}_\lambda} \tag{7}$$

where the quantum averaged energy is $\bar{H}[\boldsymbol{\gamma}](\lambda, \mathcal{B}) = \mathrm{Tr}\{\mathbf{H}(\lambda, \mathcal{B})\boldsymbol{\gamma}\}$, the classical partition function is $\mathcal{Z}_\lambda \equiv \int \mathrm{d}\boldsymbol{\gamma}\, \exp(-\beta' \bar{H}[\boldsymbol{\gamma}](\lambda; \mathcal{B}))$. The statistical temperature β' is selected so that the P and ρ ensemble have the same average energy. This partition function enables us to define the free energy change of a P-ensemble driven out of equilibrium via a change in the external drive parameter from an initial value, λ_0, to a final value, λ_f, as $\Delta F = -\beta'^{-1}\log[\mathcal{Z}_{\lambda_f}/\mathcal{Z}_{\lambda_0}]$. Notably, this maximum entropy formalism can be extended to distributions in Hilbert space by additionally constraining the distribution to appropriate subsets of Liouville space.

External Driving and Work. Consider a quantum system driven out of equilibrium by an external force. This nonequilibrium driving can be described by the Hamiltonian $\mathbf{H}(\lambda; \mathcal{B})$, where λ is an externally controllable *work parameter* and $\mathcal{B}$ represents any time-antisymmetric Hamiltonian parameters (such as the magnetic field). We assume this system is driven over an interval $[0, \tau]$ by varying λ according to a (forward) work protocol $\Lambda_{\mathrm{F}} \equiv \{\lambda_t \,|t \in [0, \tau]\}$. The resulting system can thus be described by the time-dependent Hamiltonian $\mathbf{H}_{\mathrm{F}}(t) = \mathbf{H}(\lambda_t, \mathcal{B})$. We define the reverse work protocol $\Lambda_{\mathrm{R}} \equiv \{\tilde{\lambda}_t = \lambda_{\tau-t}|t\in [0, \tau]\}$ with time-dependent Hamiltonian $\mathbf{H}_{\mathrm{R}}(t) = \boldsymbol{\Theta}\mathbf{H}(\tilde{\lambda}_t; \mathcal{B})\boldsymbol{\Theta} = \mathbf{H}(\lambda_{\tau-t}; -\mathcal{B})$, where $\boldsymbol{\Theta}$ is the time-reversal operator that inverts the sign of all time-antisymmetric observables (e.g., velocities and momenta).

Both the forward and reverse work protocols cause a net energy flux into the system. To quantify the work done under such a protocol, consider a system initially in state $\boldsymbol{\gamma}_0$ that evolves to state $\boldsymbol{\gamma}_\tau$ under protocol Λ. We define the (quantum-averaged) work performed on a system as,

$$\bar{W}[\boldsymbol{\gamma}_0; \Lambda] = -\int_0^\tau \mathrm{d}t \frac{\mathrm{d}\bar{H}[\boldsymbol{\gamma}_t; \lambda_t]}{\mathrm{d}t} = \bar{H}[\boldsymbol{\gamma}_\tau; \lambda_\tau] - \bar{H}[\boldsymbol{\gamma}_0; \lambda_0] \tag{8}$$

where $\bar{H}[\boldsymbol{\gamma}; \lambda_t] = \mathrm{Tr}\ \{H[\gamma, \lambda_t]\boldsymbol{\gamma}\}$ denotes the quantum average energy, which appears in the equilibrium distribution, eq 7. We define work in this way because it is the work that is specifically relevant to the single system fluctuations that are characterized by the correlations of eq 5. This definition of work has also been used to study fluctuations in quantum jump trajectory ensembles[42] and is consistent with the classical definition in the limit of vanishing intrinsic uncertainty. Notably, by defining work in this way we avoid many of the ambiguities that arise with "quantum work",[14,21,43−45] such as the need to perform projective measurements.

The Bochkov-Kuzovlev Functional. With these definitions, we are now in the position to derive a general response function for quantum systems. We proceed by stating the central theorem of this letter.

Theorem (Bochkov-Kuzovlev Functional). Let $\mathbf{A}$ *be an observable that transforms under time reversal as* $\Theta\mathbf{A}\Theta = \epsilon_A\mathbf{A}$ *where* $\epsilon_A = \pm 1$ *is its time-reversal parity, and let* $u(t)$ *be an*

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arbitrary smooth function. Then the forward and reverse work protocols obey the following relationship,

$$\left\langle e^{\int_0^\tau \mathrm{d}u(t)\bar{A}_F(t)} e^{-\beta\bar{W}} \right\rangle_{P_0,F} = e^{-\beta\Delta F} \left\langle e^{\epsilon_A \int_0^\tau \mathrm{d}\tilde{u}(t)\bar{A}_R(t)} \right\rangle_{P_\tau,R} \quad (9)$$

where ΔF is the free energy change associated with the forward work protocol.

The proof of this theorem follows from three primary assumptions. (1) Driven system dynamics are reversible. That is, if under the forward protocol the system evolves from state $\gamma_F(0)$ to state $\gamma_F(t)$, then under the reverse protocol the system will evolve from state $\gamma_R(0) \equiv \Theta\gamma_F(t)\Theta$ to state $\gamma_R(t) = \Theta\gamma_F(0)\Theta$. It has been established in previous studies that this reversibility applies to the dynamics of all ρ-ensembles and subensembles (and thus all single system states, γ).[19,21] (2) State space distributions (i.e., $P(\gamma)$) obey Liouville's theorem. Specifically, $P(\gamma)$ for a closed system is volume conserving under dynamical flow, and therefore the dynamical transformation $\gamma_0 \to \gamma_\tau$ has a unit Jacobian and integrals over these two variables conserves the volume element, $\int d\gamma_0 = \int d\gamma_\tau$. It has been proven previously that the Liouville theorem holds for the P-ensemble in the state space of density matrices.[26] (3) The equilibrium P-ensemble obeys a Boltzmann-like canonical distribution law as derived above in eq 7.

We proceed to prove the BK theorem for quantum P-ensembles by expanding the average in the left-hand side of eq 9,

$$\left\langle e^{\int_0^\tau \mathrm{d}u(t)\bar{A}_F(t)} e^{-\beta\bar{W}} \right\rangle_{P_0,F} = \int \mathrm{d}\gamma \left(e^{\int_0^\tau \mathrm{d}u(t)\bar{A}_F(t)} \right) (e^{-\beta\bar{W}} P_0(\gamma)) \quad (10)$$

where $P_0(\gamma)$ denotes the equilibrium distribution of the undriven system. The first term in the integrand can be alternately expressed in terms of $\bar{A}_R$ by noting that,

$$\begin{aligned} \bar{A}_F(t) &= \mathrm{Tr}\{\mathbf{A}\gamma_F(t)\} = \mathrm{Tr}\{\Theta\mathbf{A}\Theta\gamma_R(\tau - t)\} \\ &= \epsilon_A \bar{A}_R(\tau - t) \end{aligned} \quad (11)$$

where we have utilized the cyclic invariance of the trace and, in the second equality, $\gamma_F(t) = \Theta\gamma_R(\tau - t)\Theta$, which follows from time reversal symmetry of quantum dynamics. Thus,

$$\int_0^\tau \mathrm{d}tu(t)\bar{A}_F(t) = \epsilon_A \int_0^\tau \mathrm{d}t\tilde{u}(t)\bar{A}_R(t) \quad (12)$$

in which we have used the definition of the time-reversed trial function and reindexed the integral to $t \to \tau - t$. The second term in the integrand of eq 10 can be expressed as,

$$e^{-\beta\bar{W}} P_0 = \frac{e^{-\beta(\bar{W}[\gamma_0;\Lambda_F] + \bar{H}[\gamma_0;\lambda_0])}}{\mathcal{Z}_0} = P_\tau(\gamma_\tau) e^{-\beta\Delta F} \quad (13)$$

The proof of eq 9 is completed by substituting eq 12 and eq 13 into eq 10, thus yielding,

$$\begin{aligned} &e^{-\beta\Delta F} \int \mathrm{d}\gamma_0 \, e^{\epsilon_A \int_0^\tau \mathrm{d}t\tilde{u}(t)\bar{A}_R(t)} P_\tau(\gamma_\tau) \\ &= e^{-\beta\Delta F} \left\langle e^{\epsilon_A \int_0^\tau \mathrm{d}t\tilde{u}(t)\bar{A}_R(t)} \right\rangle_{P_\tau,R} \end{aligned} \quad (14)$$

where the equality follows from a trivial change in integration variable from $\gamma_0 \to \gamma_\tau$.

The Bochkov-Kuzovlev functional, eq 9 can be manipulated to derive many statistical mechanical relationships. As examples, we will use this functional to derive the Jarzynski nonequilibrium work relations and a linear response theory for quantum P-ensembles.

Deriving Nonequilibrium Work Relations. The Jarzynski relation can be derived by defining the arbitrary function $u(t) = 0$. In that case, eq 9 simplifies to a P-ensemble generalization of the Jarzynski equality,[13,15−17]

$$\langle e^{-\beta\bar{W}} \rangle_{P_0,F} = e^{-\beta\Delta F} \quad (15)$$

which relates the average work to the equilibrium free energy change over the work protocol. Using the convexity of exponentials we obtain the following useful corollary which bounds the quantum averaged work:

$$\langle \bar{W} \rangle \le \Delta F \quad (16)$$

which represents a compact statement of the second law of thermodynamics for P-ensembles.

Deriving Linear Response Theory. A P-ensemble linear response theory can be derived by combining the BK relationship of eq 9 with the CCF of eq 5. To begin this derivation, consider the case where the system couples linearly to the external forces,

$$\mathbf{H}(t) = \mathbf{H_0} - \lambda_t \mathbf{Q} \quad (17)$$

where $\mathbf{Q}$ is a collective coordinate of the system conjugate to force λ_t. Following the original classical derivation, we assume that the external force vanishes at the beginning and end of the work protocol, that is, $\lambda_0 = 0 = \lambda_\tau$, and that the system is equilibrated at the beginning and end of the forward and reverse work protocols, i.e., $P_0 = P_\tau = P$. We perform a functional derivative of eq 9 with respect to $u(t) = \tilde{u}(\tau - t)$ and take $u(t) \to 0$ to yield

$$\langle \bar{A}_F(\tau) \, e^{-\beta\bar{W}} \rangle_{P,F} = \epsilon_A \langle \bar{A}_R(0) \rangle_{P,R} \quad (18)$$

where $\Delta F = 0$ because $P(0) = P(\tau)$. The ensemble average on the right-hand side of this equation is taken at $t = 0$ of the reverse process and is therefore equivalent to an average over the time reversed quantity in the forward process, i.e., $\epsilon_A\langle \bar{A}(0)\rangle_{P_0,R} = \langle \bar{A}(\tau)\rangle_{P_0,F}$, thereby framing the entire problem in terms of the forward process. Furthermore, since the averages are evaluated at the end points of the work protocol, we can express the relationship simply in terms of equilibrium averages.

We note that $\bar{W}$ can be expressed in integral form as

$$\bar{W}[\Lambda] = -\int_0^\tau \mathrm{d}t\dot{\lambda}_t \bar{Q}[\gamma_t; \Lambda] = \int_0^\tau \mathrm{d}t\lambda_t \dot{\bar{Q}}[\gamma_t; \Lambda] \quad (19)$$

where the second equality follows from an integration by parts. Substituting this work expression into eq 18 and expanding to first order in λ_t gives,

$$\langle \bar{A}(\tau) \rangle - \langle \bar{A} \rangle_{P_{\lambda_0}} = \beta \left\langle \int_0^\tau \mathrm{d}t\lambda_t \bar{A}(\tau)\dot{\bar{Q}}(t) \right\rangle_{P_{\lambda_0}} \quad (20)$$

Noting that λ_t is a deterministic external force, we can bring the P ensemble average into the time-integral, yielding

$$\left\langle \int_0^\tau \mathrm{d}t\lambda_t \bar{A}(\tau)\dot{\bar{Q}}(t) \right\rangle_{P_0} = \int_0^\tau \mathrm{d}t\lambda_{\tau-t}\beta\langle \bar{A}(0)\dot{\bar{Q}}(t)\rangle_{P_0} \quad (21)$$

Substituting into eq 20 gives the Kubo expression for linear response,

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$$\langle \overline{A}(\tau)\rangle = \langle \overline{A}\rangle_{\text{eq}} + \int_0^\tau \mathrm{d}t \lambda_{\tau-t}\phi_{AQ}(t) \tag{22a}$$

$$\phi_{AQ}(t) = \beta C_{A\dot{Q}}(t) \tag{22b}$$

where $\phi_{AQ}(t)$ is called the linear response or after effect function. This expression characterizes the linear response of a quantum P ensemble in terms of its equilibrium CCFs. Consequently, eq 22a describes how the linear response of the ensemble is encoded in the heterogeneity between individual quantum systems at equilibrium, providing complementary information to existing ρ ensemble results describing intrinsic uncertainty. While we have restricted our attention to linear (i.e., two-index) response, the BK functional, eq 9 encodes all orders of nonlinear response which can be derived using the same methods as the classical result.[18]

We now contrast the P-ensemble BK theorm to the ρ-ensemble analog that has been developed previously in ref 19. When expressed in our notation, this theorem is given by,

$$\begin{aligned}&\overline{\left(\mathrm{e}^{\int_0^\tau \mathrm{d}tu(t)\mathbf{A}(\rho)_{\mathrm{F}}(t)}\,\mathrm{e}^{-\beta\mathbf{H}(\rho)_{\mathrm{F}}(\tau)}\,\mathrm{e}^{\beta\mathbf{H}(\rho)_{\mathrm{F}}(0)}\right)}\\&\quad= \mathrm{e}^{-\beta\Delta f}\overline{\left(\mathrm{e}^{\epsilon_A\int_0^\tau \tilde{u}(t)\mathbf{A}(\rho)_{\mathrm{R}}(t)}\right)}\end{aligned} \tag{23}$$

where $\Delta f = -\beta^{-1}\mathrm{Tr}\{\exp[-\beta\mathbf{H}_{\mathrm{F}}(\tau)]\}\mathrm{Tr}\{\exp[-\beta\mathbf{H}_{\mathrm{F}}(0)]\}$ is the ρ ensemble free energy. The subtle yet important difference between this ρ-ensemble relationship and the P-ensemble analog of eq 9, is in the performance of the quantum averages. In the ρ-ensemble expression, the entire exponential is subject to the quantum average. In this way, the ensemble heterogeneity is represented only implicitly within ρ via eq 1. In the P-ensemble expression, the quantum average appears within the exponential function, and this function itself is subject to an average over ensemble heterogeneity.

Physically, changing the order of these averages allows them to be considered separately. As a consequence, we were able to formulate a nonequilibrium work relation, eq 15, and a linear response relation, eq 22a, in terms of the quantum averaged quantities, revealing the specific role of ensemble heterogeneity in encoding nonequilibrium thermodynamics. Of particular note, the Jarzynski equality of eq 15 for quantum averaged work does not suffer from the complications of noncommuting Hamiltonians, allowing for an expression that directly references work, in contrast to the expression derived from eq 23.

Moreover, by performing the quantum average as the first step, eq 9 and it is corollaries are essentially classical statements. They construct a classical probability space on Liouville space and exploit isomorphisms between quantum and classical dynamics to characterize the classical-like heterogeneity in quantum systems. The success of this approach affirms that heterogeneity in quantum ensembles follows classical statistical mechanics, motivating future studies examining the generalization of other statistical mechanical identities to quantum P ensembles.

Example: Spin in a Magnetic Field. To illustrate the application of these P-ensemble results, we consider the simplest nontrivial example, an isolated spin system in a magnetic field. Letting the applied field $\mathbf{B}$ define the z direction, the bare Hamiltonian of this system is given by

$$\mathbf{H}_0 = -\frac{\hbar\omega_0}{2}\boldsymbol{\sigma}_z \tag{24}$$

where $\omega_0 = gB$ is the Larmor frequency induced by the applied field, g is the gyromagnetic ratio of the spin and $\boldsymbol{\sigma}_z = |0\rangle\langle 0| - |1\rangle\langle 1|$ is the Pauli z matrix.

To express the state of spin systems, it is useful to introduce the Bloch vector notation which represents a density matrix by a real valued 3D vector

$$\boldsymbol{\gamma} = \frac{1}{2}(\boldsymbol{\sigma}_0 + x\boldsymbol{\sigma}_x + y\boldsymbol{\sigma}_y + z\boldsymbol{\sigma}_z) \to (x, y, z) \tag{25}$$

where $\boldsymbol{\sigma}_0 = |0\rangle\langle 0| + |1\rangle\langle 1|$ is the identity operator and the remaining Pauli matrices are given by $\boldsymbol{\sigma}_x = |0\rangle\langle 1| + |1\rangle\langle 0|$ and $\boldsymbol{\sigma}_y = i|0\rangle\langle 1| - i|1\rangle\langle 0|$. Using the quantum averaged notation defined above, the Bloch vector components can be conveniently written as $j = \overline{\sigma_j}[\boldsymbol{\gamma}]$ for $j = x, y, z$. That is, they give the components of the systems quantum averaged spin angular momentum.

The ensemble averaged equilibrium density matrix is given by

$$\boldsymbol{\rho} = \frac{\mathrm{e}^{-\beta\boldsymbol{H}_0}}{\mathrm{Tr}\,\mathrm{e}^{-\beta\boldsymbol{H}_0}} \to -\left(0, 0, \tanh\left(\frac{\beta\hbar\omega_0}{2}\right)\right) \tag{26}$$

while the equilibrium P-ensemble distribution is given by

$$P(\boldsymbol{\gamma}) = \frac{\mathrm{e}^{-\beta'\hbar\omega_0 z/2}}{Z_0} \tag{27}$$

where we have used the notation $z = \overline{\sigma_z}[\boldsymbol{\gamma}]$, and $Z_0 = \int \mathrm{d}\boldsymbol{\gamma}\,\exp\left(-\frac{\beta'\hbar\omega_0}{2}z\right)$ is the P ensemble partition function. Evaluating the $\boldsymbol{\gamma}$ integral over the allowed spin states in this partition function is straightforward. Since we are considering an ensemble of isolated spin systems with no environment, each individual system must be found in a pure state. In the Bloch sphere representation this corresponds to vectors of unit length, and the $\boldsymbol{\gamma}$ integral is simply an integral over the surface of the unit sphere, giving

$$\begin{aligned}Z_0 &= \int \mathrm{d}\boldsymbol{\gamma}\,\mathrm{e}^{-\beta'\hbar\omega_0/2z}\\&= \int_{-\pi}^{\pi}\mathrm{d}\phi\int_0^\pi \mathrm{d}\,\theta\,\sin(\theta)\,\mathrm{e}^{-\beta'\hbar\omega_0/2\cos(\theta)}\\&= \frac{8\pi\,\sinh\left(\frac{\hbar\beta'\omega_0}{2}\right)}{\hbar\beta'\omega_0}\end{aligned} \tag{28}$$

By requiring that eqs 26 and 28 satisfy the consistency constraint eq 1, it is straightforward to see that the x and y components are satisfied for any choice of β and β'. All that remains is therefore to compute $\langle z\rangle$ from eq 27 and equate it to $\tanh(\hbar\beta\omega_0/2)$ from eq 26 to give

$$\beta = \frac{2}{\hbar\omega_0}\,\mathrm{arctanh}\left(\frac{2}{\hbar\omega_0\beta'} - \frac{1}{2}\coth\left(\frac{\hbar\beta'\omega_0}{2}\right)\right) \tag{29}$$

We now consider the response of the system to a time-dependent external field λ_t along an axis defined by unit vector $\boldsymbol{q} = (q_x, q_y, q_z)$. This leads to a perturbing Hamiltonian

$$\boldsymbol{h}(t) = -\lambda_t\boldsymbol{\sigma}_{\boldsymbol{q}} \tag{30}$$

where $\boldsymbol{\sigma}_{\boldsymbol{q}} = q_x\boldsymbol{\sigma}_x + q_y\boldsymbol{\sigma}_y + q_z\boldsymbol{\sigma}_z$ and we have taken λ_t to have units of energy for simplicity of notation. Using the notation of eq 17, $\mathbf{Q} = \boldsymbol{\sigma}_{\boldsymbol{q}}$. We will now evaluate the linear response function for this perturbation using both the P-ensemble

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expression in eq 22a and the traditional ρ-ensemble expression. Any operator of the spin system can be written in the form $\mathbf{A} = A_0\,\boldsymbol{\sigma}_0 + A\boldsymbol{\sigma}_a$ of a constant shift A_0 plus a spin operator along an axis $\boldsymbol{a}$. It therefore suffices to consider responses of spin operators along an arbitrary axis (i.e., letting $\mathbf{A} = \boldsymbol{\sigma}_a$). At the outset we note that both approaches should give the same result since $\langle \overline{A}\rangle = \overline{A}[\boldsymbol{\rho}]$ indicating that the P and ρ linear response theories are predicting the same quantities in terms of different correlation functions.

We can first evaluate the response of operator $\mathbf{A}$ using the traditional ρ-ensemble expression first derived by Kubo[4] in the same form as eq 22a but with the linear response function

$$\phi_{AQ}^{(\rho)}(t) = \overline{[\mathbf{A}, \mathbf{Q}(t)]}[\boldsymbol{\rho}] \tag{31}$$

in terms of ρ-ensemble commutators in the Heisenberg picture. To evaluate this expression it is useful to write down the Heisenberg picture spin operators

$$\begin{aligned}\boldsymbol{\sigma}_a(t) = (&\cos(\omega_0 t)\mathbf{a} + \sin(\omega_0 t)(\mathbf{b}\times\mathbf{a})\\ &+ (\mathbf{a}\cdot\mathbf{b})(1-\cos(\omega_0 t))\mathbf{b})\cdot\vec{\boldsymbol{\sigma}}\end{aligned} \tag{32}$$

where $\mathbf{b} = (0, 0, 1)$ is the magnetic field unit vector and the Pauli vector $\vec{\boldsymbol{\sigma}} = (\boldsymbol{\sigma}_x, \boldsymbol{\sigma}_y, \boldsymbol{\sigma}_z)$ notation allows for a compact expression in terms of vector dot products $(\cdot)$ and cross products $(\times)$. Substituting eq 32 into eq 31 and evaluating the trace yields the ρ-ensemble linear response function

$$\begin{aligned}\phi_{AQ}^{(\rho)}(t) = \frac{2}{\hbar\omega_0}\tanh\!\left(\frac{\hbar\beta\omega_0}{2}\right)&[\cos(\omega_0 t)(\mathbf{b}\cdot(\mathbf{q}\times\mathbf{a}))\\ &+ \sin(\omega_0 t)((\mathbf{q}\cdot\mathbf{a}) - (\mathbf{q}\cdot\mathbf{b})(\mathbf{a}\cdot\mathbf{b})]\end{aligned} \tag{33}$$

Consider now the P-ensemble linear response expression, eq 22b. To evaluate this expression, we need to evaluate the quantum averaged spin operators $\overline{\sigma_a}[\boldsymbol{\gamma}]$ and their time derivatives. Using the Heisenberg picture operators in eq 32 these can be evaluated to give

$$\begin{aligned}\overline{\sigma_a}[\boldsymbol{\gamma}(t)] &= \mathrm{Tr}\{\boldsymbol{\sigma}_a(t)\boldsymbol{\gamma}\}\\ &= \cos(\omega_0 t)(\boldsymbol{\gamma}\cdot\mathbf{a}) + \sin(\omega_0 t)(\boldsymbol{\gamma}\cdot(\mathbf{b}\times\mathbf{a}))\\ &\quad + (1-\cos(\omega_0)(t)(\mathbf{a}\cdot\mathbf{b})(\boldsymbol{\gamma}\cdot\mathbf{b})\end{aligned} \tag{34a}$$

$$\begin{aligned}\dot{\overline{\sigma}}_a\boldsymbol{\gamma}(t)] = \omega_0(&-\sin(\omega_0 t)(\boldsymbol{\gamma}\cdot\mathbf{a}) + \cos(\omega_0 t)(\boldsymbol{\gamma}\cdot(\mathbf{b}\times\mathbf{a}))\\ &+ \sin(\omega_0 t)(\mathbf{a}\cdot\mathbf{b})(\boldsymbol{\gamma}\cdot\mathbf{b}))\end{aligned} \tag{34b}$$

$$\begin{aligned}\overline{\sigma_a}[\boldsymbol{\gamma}(0)]\dot{\overline{\sigma}}_q[\boldsymbol{\gamma}(t)] = \omega_0(\boldsymbol{\gamma}\cdot\mathbf{a})(&\sin(\omega_0 t)((\mathbf{q}\cdot\mathbf{b})(\boldsymbol{\gamma}\cdot\mathbf{b}) - \boldsymbol{\gamma}\cdot\mathbf{a})\\ &+ \cos(\omega_0 t)(\boldsymbol{\gamma}\cdot(\mathbf{b}\times\mathbf{q})))\end{aligned} \tag{34c}$$

Equation 34b was obtained by taking the time derivative of eq 34a.

Now all that must be done is evaluating the average of eq 34c with respect to the $P(\boldsymbol{\gamma})$ from eq 27. This expression is quadratic in $\boldsymbol{\gamma} = (x, y, z)$; therefore, it is helpful to note that we can write the distribution in the product form $P(\boldsymbol{\gamma}) = P(x)P(y)P(z)$ indicating that the components of $\boldsymbol{\gamma}$ are uncorrelated (i.e., $\langle xy\rangle = \langle yz\rangle = \langle xy\rangle = 0$). Moreover, since $P(x) = P(y)$, we have $\langle x^2\rangle = \langle y^2\rangle$. By directly integrating eq 27 we obtain the expression

$$\langle x^2\rangle = \langle y^2\rangle = -\frac{2}{\hbar\omega_0\beta'}\langle z\rangle = \frac{2}{\hbar\beta'\omega_0}\tanh\!\left(\frac{\hbar\beta\omega_0}{2}\right) \tag{35}$$

Using these identities, we can compute the P ensemble response function which gives

$$\begin{aligned}\phi_{AQ}(t) &= \beta'\langle x^2\rangle[\cos(\omega_0 t)(\mathbf{b}\cdot(\mathbf{q}\times\mathbf{a})) + \sin(\omega_0 t)\\ &\qquad(\mathbf{q}\cdot\mathbf{a} - (\mathbf{q}\cdot\mathbf{b})(\mathbf{a}\cdot\mathbf{b}))]\\ &= \phi_{AQ}^{(\rho)}(t)\end{aligned} \tag{36}$$

demonstrating the equivalence between the responses computed from both P-ensemble and ρ-ensemble approaches.

The P-ensemble BK theorem therefore provides an alternative route to computing the linear response purely in terms of the classical heterogeneity between individual quantum systems. Since this expression is built on a classical ensemble, it opens the door to applying tools for computing classical responses directly to the quantum regime. Mathematically, this can be seen in the fact that the resulting response function contains neither a commutator nor a Kubo-transformed correlation function and therefore need not be directly concerned with the peculiarity of quantum observables. Moreover, this agreement is generic and not specific to the simple spin example considered here.

From a physical perspective, the agreement between these two formalisms is remarkable. While they both express linear response purely in terms of equilibrium statistics, the mechanism behind the statistics in the two expressions is very different. The ρ-ensemble response function, eq 31, arises from the noncommutativity of two observables—a manifestly quantum phenomenon. Specifically, it quantifies how much the measurement of one observable changes the statistics of another at a later time. In contrast, the P-ensemble response function, eq 22a, never involves a multitime measurement process. In fact, it is only concerned with the heterogeneity between the quantum-averaged values (single time observables $\overline{A}(t)$) in different systems in an ensemble—a feature absent in the ρ-ensemble description.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jpclett.1c01315.

Equilibrium quantum distributions calculations (PDF)

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Notes

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